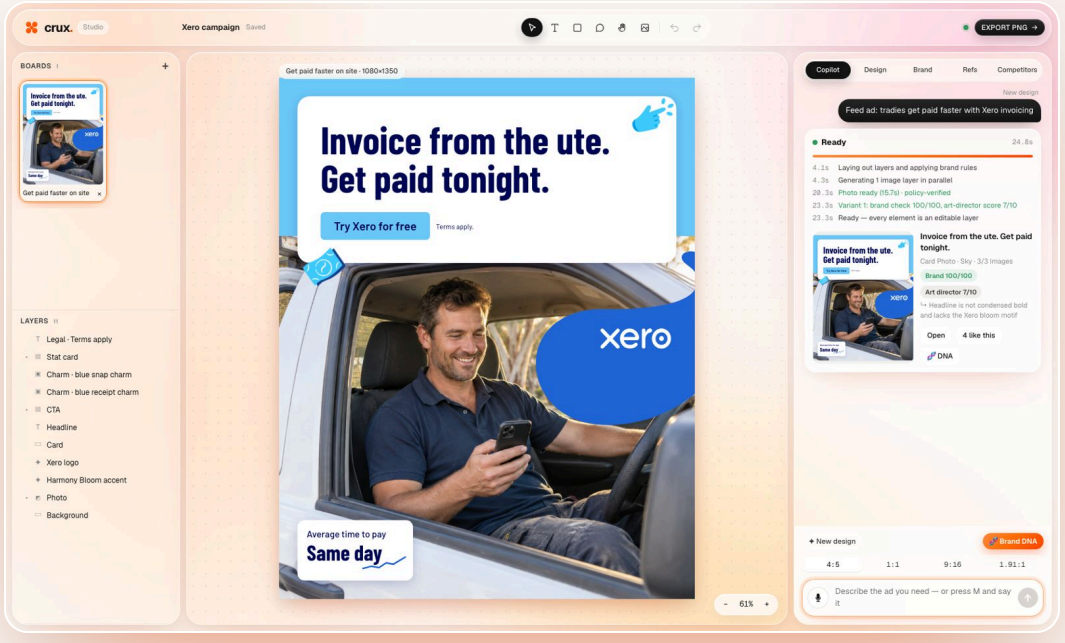
TECHNICAL PRESENTATION · GETCRUX HACKATHON

CRUX STUDIO
AI DESIGN COPILOT

A layered image editor where an AI copilot creates, edits and audits on-brand ads — by chat, voice or a comment on the canvas — orchestrated by Temporal and parallelised by Celery.

Temporal · Celery + Redis · FastAPI · React + SVG · DeepSeek V4 · GPT Image Flare · ElevenLabs Scribe v2



THE PROBLEM

BRANDED DESIGN IS SLOW — AND EASY TO GET WRONG

Today

- Designers hand-build every ad in an image editor
- Fonts, colours, logo rules and approved assets must be remembered
- Legal rules (claims, “Terms apply.”) are easy to miss
- Exploring variations or competitor ideas costs hours

Crux Studio

- Describe an ad → a layered, editable board in ~20 s
- Edit any single layer by chat, **voice** or a canvas comment
- Every output checked against brand & legal policy — and auto-fixed
- Variations, Brand DNA audits and competitor intel built in

~20s
prompt → layered ad

10+
editable layers per design

323
brand assets analysed by AI

~3/255
canvas vs export pixel difference

ARCHITECTURE

WHO DOES WHAT

Browser — Crux Studio
React + SVG canvas
layers · panels · copilot chat
voice capture · landing page

⇄

FastAPI
HTTP API · documents
uploads · render/export
voice proxy · job status

→

Temporal
durable workflows:
design · edit · Brand DNA
competitors · brand ingest

→

Celery workers + Redis
parallel heavy work:
image layers · compositing
vision checks · scraping

DeepSeek V4 Pro
plans briefs & copy, plans edits, synthesises insights

DeepSeek V4.1 Flash
vision: asset tagging, image compliance, QA, Brand DNA, ad reading

GPT Image 2.5 Flare
photos, transparent cut-outs, brand-style art, in-place photo edits

ElevenLabs Scribe v2
speech → text for voice editing (server-side proxy)

Storage
`data/` — documents (JSON), generated layers, exports, brand catalog

Brand kit
Candidate-resources → vectors, charms, tags, insights

Meta Ad Library
public competitor ads (headless Chromium)

TECH STACK

EACH TECHNOLOGY HAS ONE JOB

React + TypeScript + SVG	The editor. Draws each board as native SVG from the same layer model and font files as the server, so the canvas, previews and exports match.
FastAPI (Python)	The front door: documents, uploads, server render, starts jobs and reports their progress, proxies voice so API keys never reach the browser.
Temporal	The “project manager with a memory”: runs each multi-step job as a durable workflow — retries, timeouts, live progress, survives restarts.
Celery + Redis	The “workforce”: a pool of workers that runs heavy, slow tasks in parallel (every image layer at once). Redis is the queue between them.
DeepSeek V4 Pro / V4.1 Flash	Pro = thinking (copy, layout choices, edit plans, strategy). Flash = seeing (vision QA, compliance, tagging, ad analysis) — fast.
GPT Image 2.5 Flare	Makes pixels: photos, transparent cut-outs, brand-style illustrations/charms from reference assets, and identity-preserving photo edits.
ElevenLabs Scribe v2	Turns spoken edits into text (auto-stops on silence in the browser).
Pillow · potrace · Playwright	Server compositor & text metrics · brand shapes to vectors · Meta Ad Library scraping and browser tests.

ORCHESTRATION

TEMPORAL IS THE BRAIN. CELERY IS THE MUSCLE.

Temporal — decides and remembers

- Each job is a **workflow**: a recipe of steps (plan → layout → generate → audit)
- Remembers where it is: if a worker crashes, it resumes from the last finished step
- Retries failed steps with back-off; timeouts & heartbeats catch hangs
- Exposes a live `progress` query → the UI shows boards as soon as the layout exists
- Runs steps concurrently (4 variants, 20 ad analyses) with deadlines

Celery — does the heavy lifting

- A pool of 24 worker threads pulling tasks from a Redis queue
- One task per image layer → all layers of all variants render at the same time
- Also: compositing + brand audit, AI vision checks, Ad Library scraping
- Scale out by adding workers — no workflow code changes
- Temporal activities hand a task to Celery and await it without blocking

How they meet

Workflow step → `celery_task` activity sends the task to Redis → a Celery worker runs it → the activity polls the result (heartbeating) → Temporal records it and moves to the next step. Durability from Temporal, throughput from Celery.

IMAGE GENERATION

FROM A SENTENCE TO A LAYERED AD

1 · Plan

DeepSeek Pro writes the brief: archetype, colourway, headline, CTA, which brand assets, photo prompt — from brand rules, voice & references. Legal re-plan if needed.

`~4s · Temporal activity`

→

2 · Lay out

Code (not the AI) places every layer: text measured with the real font files, safe zones, waves kept off the copy. Brand guard runs.

`<1s · deterministic`

→

3 · Generate

Each image layer becomes a Celery task: GPT Image Flare photo / cut-out / brand-style art — all in parallel.

`~11-15s · Celery`

→

4 · Verify

Vision AI checks each image for text, logos, watermarks, competitor branding → regenerates with a correction if it fails.`~2-3s / image`

→

5 · Compose & audit

Server compositor flattens the layers; brand guard auto-fixes (contrast, logo colour, “Terms apply.”); art-director QA scores it.`~3s · Celery`

Progressive canvas

The board appears in the editor right after step 2 with placeholders; each photo drops in the moment its Celery task finishes. Total ≈ 20–25 s.

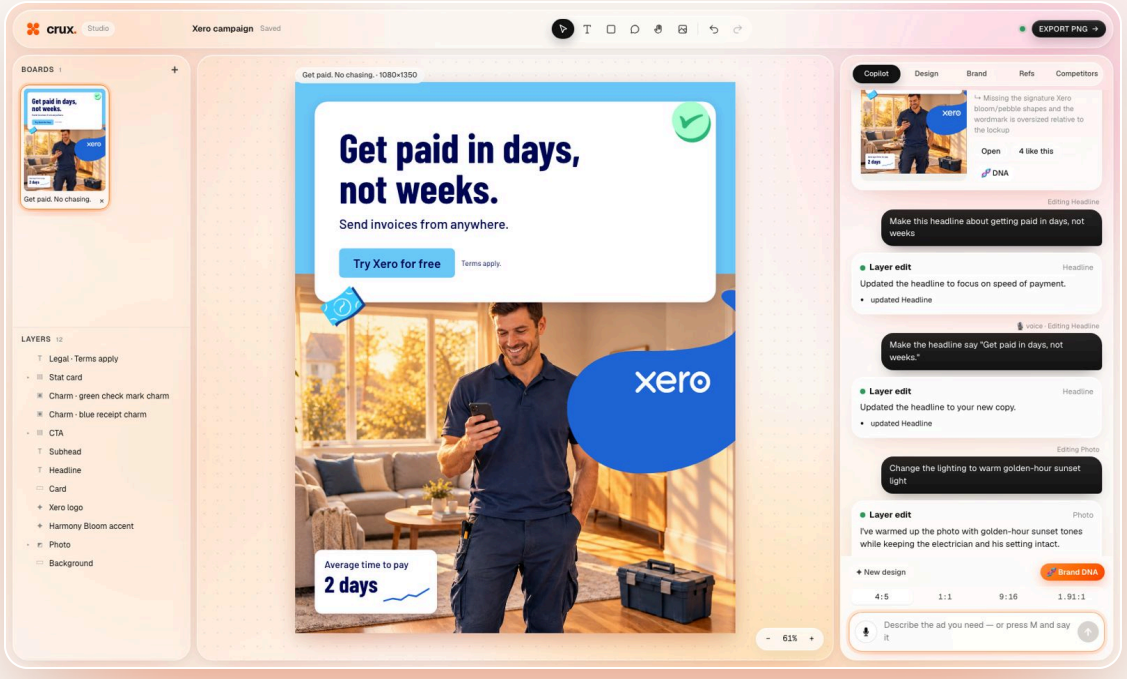
Why layers, not one picture

Only photos/illustrations are generated images. Headline, CTA, card, bloom, logo and charms are real vector/text/brand-asset layers — so every element stays editable and on-brand.

LAYERED DESIGN MODEL

ONE MODEL, THREE IDENTICAL VIEWS

- `DesignSpec` = board + ordered layers: rect, text, image, vector, group, mask
- Editor canvas draws it as SVG; the server compositor (Pillow) draws it as PNG
- Same font files & ascent/descent metrics, same wrap rules, same masks and shadows
- Measured in tests: canvas vs server render differ by ~2–4 / 255 per pixel
- Photos sit inside bloom masks; logo & blooms are traced vectors (recolourable)
- Documents autosave as JSON; undo/redo, locks, visibility, reorder



BRAND KIT INGESTION

TEACHING THE AI THE BRAND

Scan
Candidate-resources: logo, blooms, background templates, illustrations, charms, product UI, 20 reference ads.

→

Vectorise
potrace turns the wordmark, 12 named blooms and 18 wave templates into clean, recolourable SVG paths.

→

Slice
The charm sprite sheet is cut into 251 individual charms (connected-component analysis).

→

Tag with vision
Flash names & tags every asset in numbered contact sheets; deconstructs each reference ad (archetype, copy angle).

→

Synthesise
Pro distils brand voice, headline formulas, CTA bank and layout patterns → `catalog.json`.

323

assets catalogued (Temporal BrandIngestWorkflow, ~45s, cached by fingerprint)

5

layout archetypes learned from the references: card+photo, photo in bloom, navy illustration, device, checklist

11

palette colours — primary for large fills, accents only for charms

EDITING

THREE WAYS TO CHANGE ONE LAYER

💬 Chat

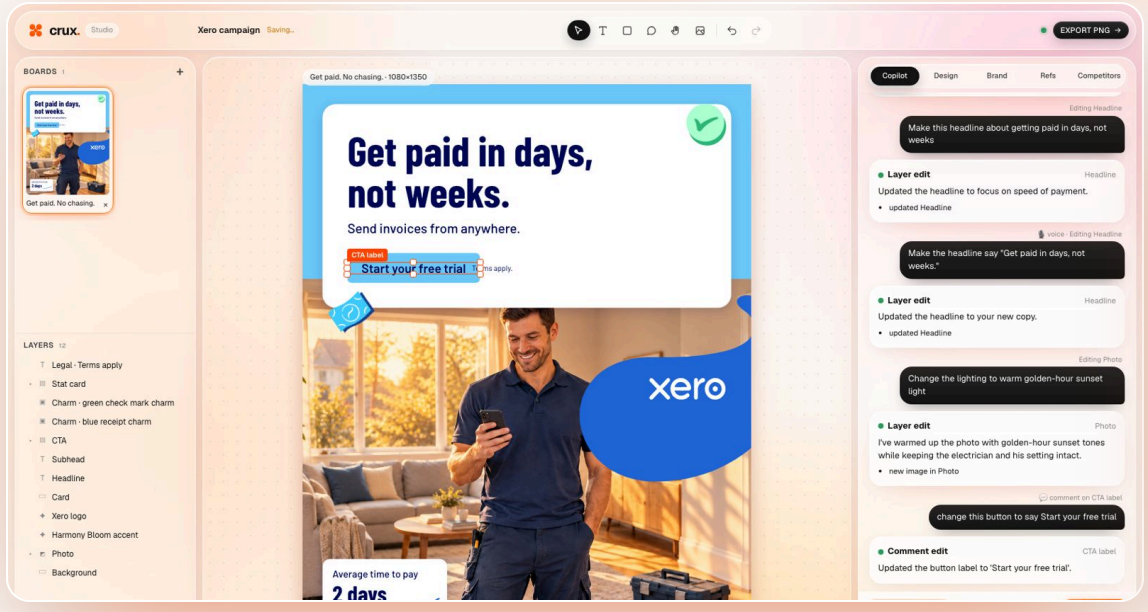
Select a layer → “make the photo golden hour”. EditWorkflow: Pro plans intent ops → compiler re-fits text with real font metrics, snaps colours, resizes CTA buttons → image edits run in Celery (same person & scene) → ops apply to the document. **~3–8s text, ~20s photo**

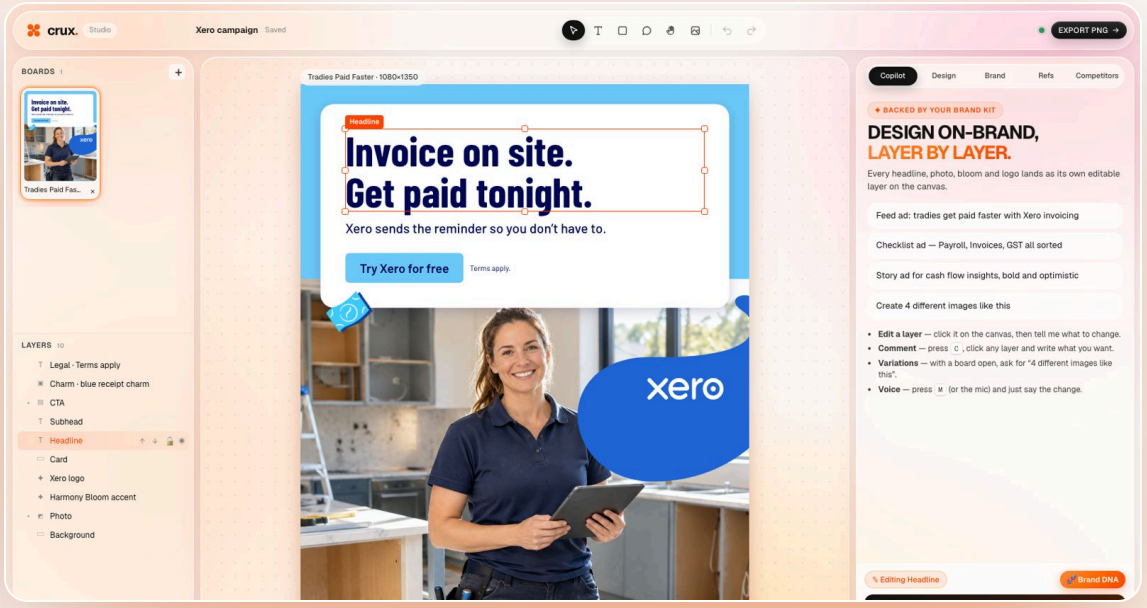
📍 Comment

Comment tool → click anywhere. The layer under the pointer is hit-tested (deep-selects into groups and masked photos), the edit runs, Crux replies in the thread and resolves it. **~3s**

🎙️ Voice

Press M → browser records (MediaRecorder) with a live level meter, auto-stops on 1.3s of silence → FastAPI proxies the audio to ElevenLabs Scribe v2 → the transcript runs as a chat edit. **~8s incl. speaking**

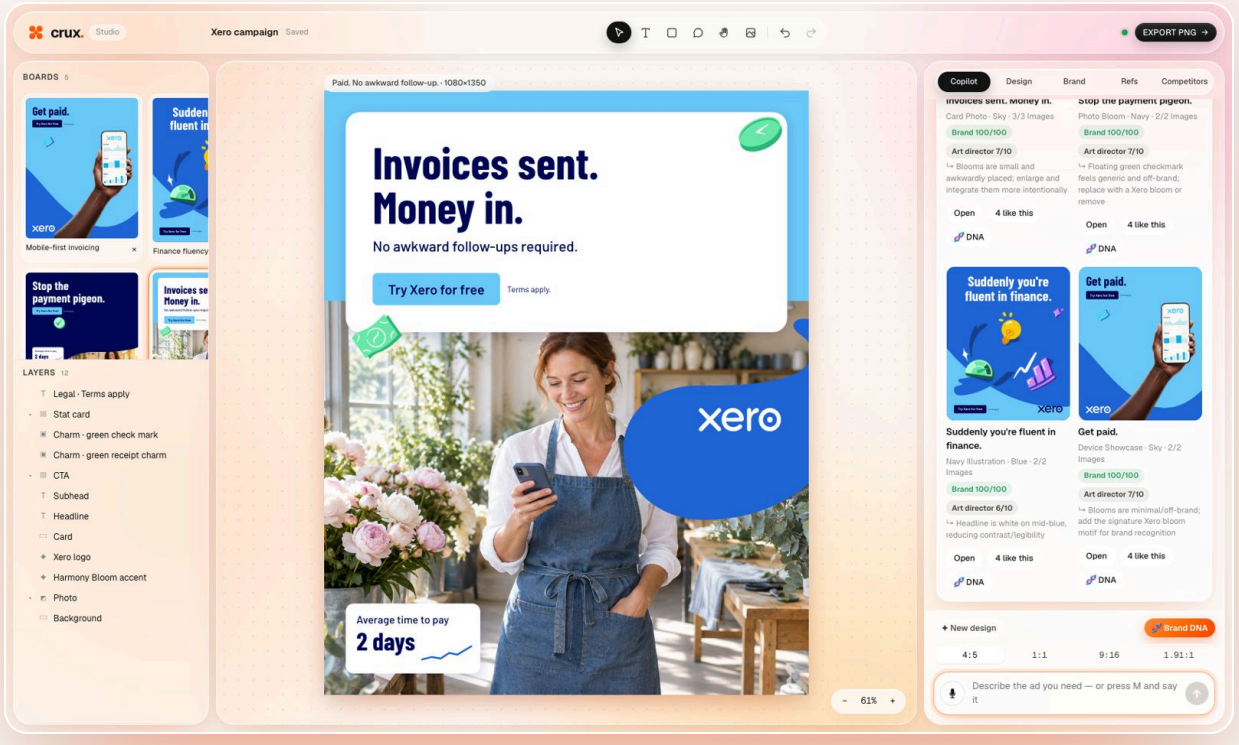




GENERATE LIKE THIS

1–4 REAL ALTERNATIVES, IN PARALLEL

- Pick a board (or a reference / competitor ad) → choose how many
- Diversity is enforced in code: each variant gets a different archetype; one gets a contrasting colourway
- Temporal plans all variants concurrently; Celery renders every image layer of every variant at once
- 4 variants ≈ the time of one (~23–27s)
- Non-Crux boards: rendered and read by vision AI as the reference



ENTERPRISE GUARDRAILS

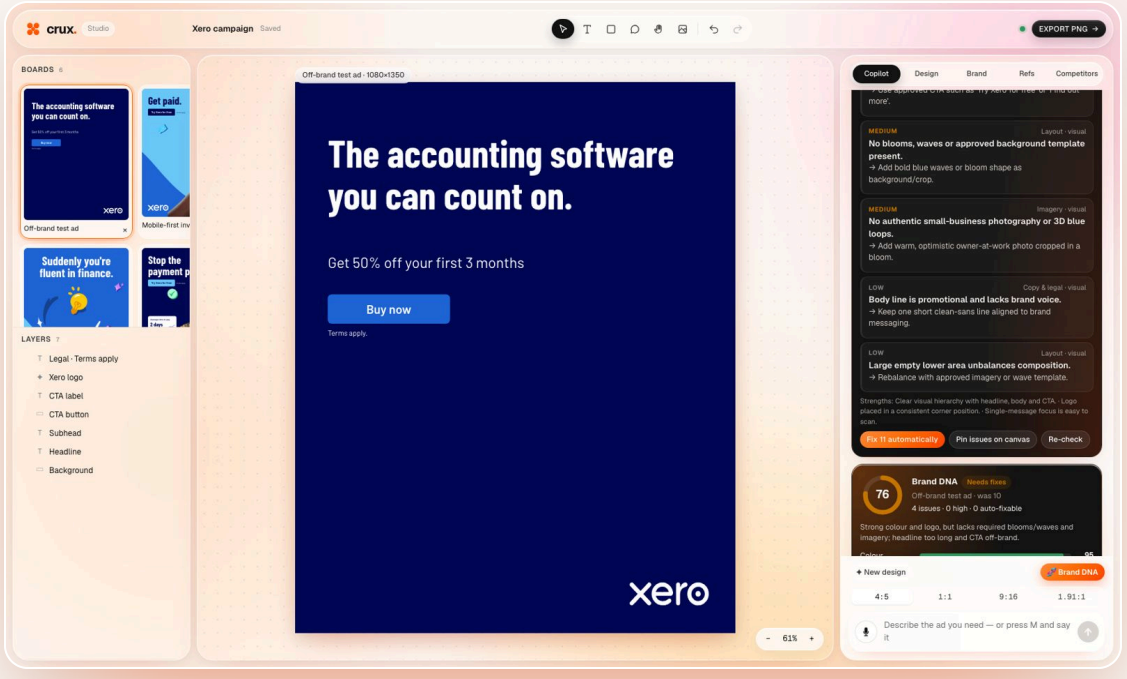
BRAND & LEGAL POLICY AT EVERY STAGE

Planning	Unapproved claims (guaranteed, best, #1, risk-free, free forever, 100%…) and competitor names are rejected → the brief is re-planned with legal feedback.
Layout	Brand fonts only, measured to fit; story safe zones; logo in approved colourways with clear space; “Terms apply.” added for any offer.
Images	Every AI image is vision-checked for text, logos, watermarks, competitor branding, unsafe content (and off-palette art) → regenerated up to 3× with a targeted correction.
Composite	Brand guard samples the real pixels behind each text/logo: WCAG contrast ≥ 4.5:1 and logo colour are auto-fixed; palette snapping.
Editing	Off-policy requests (remove/recolour/rotate logo, unapproved claims) are blocked or turned into the closest compliant change, with an explanation.
Resilience	Reviewer AI calls have short timeouts — if a reviewer is down, delivery continues and the image is marked “unverified”.

BRAND DNA

AUDIT ANY DESIGN. FIX IT IN ONE CLICK.

- **Measured audit** (deterministic): fonts, palette, contrast against the post-fix background, logo colour/size/clear space, legal copy with compliant AI rewrites, safe zones
- **Visual audit** (Flash vision) grounded in the measured facts — imagery, layout, tone
- Score across 6 dimensions; every issue located (bounding box) and explained
- Fix all → re-check (off-brand test ad: ~10 → ~60–75); pins on the canvas
- BrandDnaWorkflow on Temporal, ~10s



COMPETITOR INTEL

LEARNING FROM ADS THAT KEPT RUNNING

Scrape
Celery task drives headless Chromium on the public Meta Ad Library (no login): ~20 ads — creative, copy, advertiser, "started running on".~6s

→

Rank
Days active = performance proxy (winners stay live). Tiers: proven ≥ 90d · testing · new. Creatives cached locally.

→

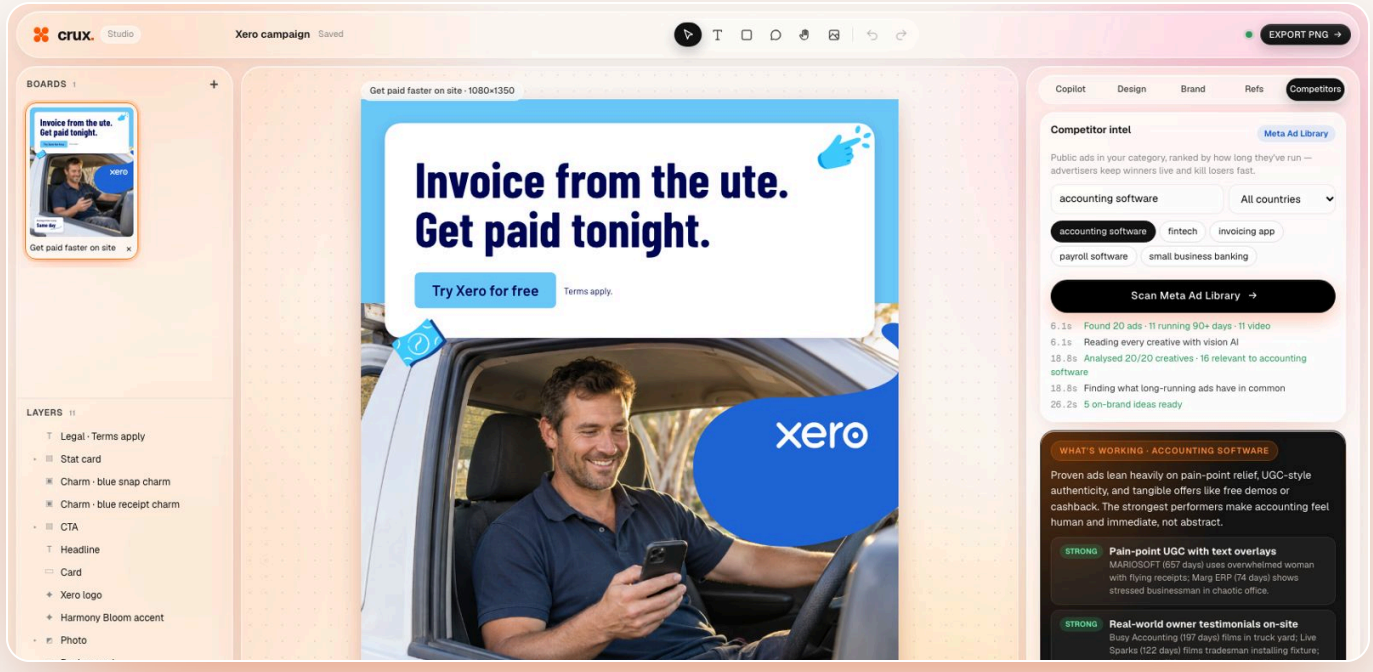
Read
Flash vision reads every creative in parallel: hook, angle, format, relevance — with a 20s deadline.~5-20s

→

Synthesise
Pro weights patterns by longevity, finds whitespace and writes 5 original Xero ideas (legal-checked).~10s

→

Create
"Create this →" or "Remix on-brand" runs the normal DesignWorkflow.



Measured

- 20 ads, all dated, all with creative
- Longest-running: 657 days
- Scan 21–36s end to end
- Ideas never reuse competitor copy, claims or branding

PERFORMANCE & RELIABILITY

MEASURED, NOT PROMISED

~22s

new layered ad (plan 4s · images in parallel ~12s · compose + QA 3s)

~3s

layer text edit / comment edit

~25s

4 variations (same as one — parallel)

~8s

voice edit, incl. speaking & transcription

Durable

Every job is a Temporal workflow: retries with back-off, timeouts & heartbeats; one failed layer degrades to a placeholder instead of failing the design.

Fail fast where advisory

Compliance/QA reviewers time out in 15–25s; analysis deadlines cap slow vision calls; LLM rate-limits retried with jitter, Pro ↔ Flash fallback.

Fast by design

Geometry is computed in code (no LLM round-trips per element); progressive canvas shows boards in ~4s; brand catalog cached by fingerprint.

QUALITY & SECURITY

VERIFIED IN A REAL BROWSER

Automated tests

- `scripts/smoke_test.py` — API: 4 formats, variations, edits
- `tests/e2e_studio.py` — Playwright in Chromium: create, every image loaded, canvas-vs-server pixel diff, layer / photo / comment / voice edits (fake microphone), drag + undo + inline text, variations, Brand DNA fix-all, live Ad Library scan → idea → board
- Throwaway documents per run — the designer's work is never touched

Security

- API keys only in `.env` (git-ignored, mode 600), loaded as secrets, never logged or sent to the browser
- Voice audio proxied server-side; Ad Library scraped with one gentle session
- Services bind to 127.0.0.1; CORS limited; uploads size-capped & validated; ids and file paths validated

RUN IT

ONE COMMAND

```
cp .env.example .env  # add keys
./run.sh              # start everything
./run.sh test     # smoke + browser E2E
./run.sh stop
```

Starts Redis, Temporal dev server, Celery workers, the Temporal worker and FastAPI; builds the editor; analyses the brand kit on first run. Landing: `localhost:8787` · Studio: `/studio`

Repository map

- `backend/crux/brand` — kit, ingestion, policy, guard, compliance, Brand DNA
- `backend/crux/design` — spec, layout engine, compositor, planner, editor
- `backend/crux/research` — Meta Ad Library + competitor analysis
- `backend/crux/workflows` · `tasks` · `api` — Temporal, Celery, FastAPI
- `editor/src` — model, SVG renderer, canvas, panels, copilot, voice, landing

CRUX STUDIO

DESIGN ON-BRAND, LAYER BY LAYER.

Temporal keeps every job durable. Celery makes every layer parallel. The AI writes, sees and checks — and the code keeps every pixel on-brand and editable.

